

A RAG-Based Architecture for Automating ESRS E1 Climate Disclosure Alignment

Yajas Kandel

March 12, 2026

1 Introduction

The transition from voluntary sustainability disclosure to mandatory, audit-grade reporting represents one of the most significant regulatory shifts in the history of European corporate law. The Corporate Sustainability Reporting Directive (CSRD) has fundamentally altered the compliance landscape, requiring over 50,000 companies to provide granular disclosures on their environmental, social, and governance (ESG) impacts. At the heart of this directive are the European Sustainability Reporting Standards (ESRS), a multi-hundred-page framework designed to ensure "double materiality"—the concept that companies must report both on their financial risks from climate change and their external impact on the planet.

However, the implementation of these standards has revealed a critical "compliance gap." As noted by Arner et al. [2], the inherent complexity and "conceptual ambiguity" of new regulatory frameworks often create significant barriers to entry, particularly for small-to-medium enterprises (SMEs) that lack the resources of large multinational corporations. Organizations currently face the daunting task of translating high-level legal requirements into operational data collection workflows. This manual process is not only slow and costly but also susceptible to "greenwashing", which is the practice of providing unstructured, non-verifiable sustainability claims that mask a lack of actual performance [3].

Within the field of Computer Science, Regulatory Technology (RegTech) has emerged as a promising field to address these inefficiencies. Specifically, Retrieval-Augmented Generation (RAG) offers a specialized solution for navigating "knowledge-intensive" domains like climate law. Unlike standard Large Language Models (LLMs), which are trained on static data and are prone to "hallucinating" plausible but incorrect information, a RAG-based architecture dynamically retrieves authoritative, up-to-date documents during the generation process [6].

The primary objective of this project is to develop a RAG-based architecture for navigating ESRS E1 (Climate Change) disclosures. The ultimate goal is to see if a specialized AI tool can be designed to support sustainability teams in aligning their internal data with the ESRS E1 standard. The AI will function as a "Knowledge Assistant" that provides evidence based answers to reporting queries by retrieving relevant passages from the ESRS E1 and the relevant European Financial Reporting Advisory Group (EFRAG) guidelines, generating grounded explanations through a RAG pipeline. The system will help organizations interpret disclosure requirements, identify necessary data inputs, and reduce the workload

for report writing teams by offering transparent, traceable, and regulation-aligned guidance. Although the scope of this study is limited to the ESRS E1 standard, through continued system updates to cover all ESRS standards from E1-E9, a tool like this may someday be used as a full-fledged commercial product that assists with CSRD compliance all over Europe.

2 Background

To understand the architecture of this system designed for automated regulatory alignment, one must first understand the two foundational pillars of the project: the technical mechanics of Retrieval-Augmented Generation (RAG) and the structural complexity of the European Sustainability Reporting Standards (ESRS).

2.1 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an architectural framework designed to enhance the capabilities of Large Language Models (LLMs) by providing them with access to external, authoritative knowledge bases. As established by Lewis et al. [6], RAG combines the "parametric memory" of pre-trained models with a "non-parametric memory" consisting of a dense document index. Parametric memory refers to the memory of a Large Language Model itself, in which it has learned patterns, grammar, and general facts. This knowledge is stored in the model's parameters. The issue is that parametric memory alone is static, meaning once training is over the model does not know anything new. Non-Parametric memory refers to the external knowledge fed to the LLM through a vector database. The benefit is that this type of knowledge is dynamic and controllable, with the AI's knowledge continuously updating through what it is fed.

The process begins with Vector Embeddings, where text segments are transformed into high-dimensional numerical vectors. This project utilizes the all-MiniLM-L6-v2 model to map text into a 384-dimensional vector space. The core advantage of this approach over traditional keyword search is the ability to perform Semantic Retrieval. By calculating the Cosine Similarity between a user's query vector and the document vectors stored in a Vector Database (such as ChromaDB), the system identifies information based on conceptual meaning rather than exact word matches. This is particularly effective in high-stakes domains where Shuster et al. [7] demonstrate that providing explicit context significantly reduces the frequency of model hallucinations.

2.2 The ESRS Regulatory Framework

The domain of this project is defined by the ESRS E1 Climate Change standard. For a computational system to effectively navigate this standard, it must account for its hierarchical structure. The ESRS is not a single narrative but is partitioned into specific Disclosure Requirements (DRs) and Application Requirements (ARs).

The standard utilizes "Double Materiality" as a guiding principle, requiring companies to report on both their financial risks and their external environmental impacts. Furthermore, sustainability reports often suffer from a lack of standardization, which makes the strict

structure of the EFRAG guidelines essential for creating verifiable data. By mapping these legal sections into the system’s metadata, the RAG pipeline can provide ”source receipts”, which are traceable references back to the original legal text. This is a critical requirement for audit-grade reporting and for avoiding the risks of ”greenwashing” identified in recent sustainability and RegTech research [3, 5].

2.3 Sustainability Implications of the Technology

A unique consideration for this project is the intersection of AI performance and its own environmental impact. While massive models are often prioritized for accuracy, research by Kozhipuram et al. [8] suggests that smaller, RAG-enabled models can achieve high performance on knowledge-intensive tasks with greater efficiency. This aligns with the broader goals of ”Sustainable Artificial Intelligence” and carbon-aware optimization, which advocate for linking the development of Artificial Intelligence with sustainability driven power consumption objectives [1]. By opting for an efficient RAG pipeline over a resource-heavy baseline LLM, the system adheres to the very principles of sustainability it is designed to monitor.

3 Related Works

The development of a specialized retrieval system for ESRS E1 alignment sits at the intersection of regulatory informatics, natural language processing, and sustainable computing. This section synthesizes recent research to demonstrate how RAG and structured data extraction can address the complexities of corporate sustainability reporting.

3.1 Verifying Regulatory Alignment via RAG

The most direct precedent for this research is the work of Forster et al. (2025), who developed a machine learning framework to track ESG indicators across 600 of the largest European corporations [5]. Their research specifically highlights a ”transparency gap” where companies with high ESG ratings often provide significantly more granular data than their lower-rated peers. By utilizing a RAG pipeline to process over 9,200 corporate reports, Forster et al. demonstrated that retrieval-based models can systematically identify missing disclosures mandated by the newly established ESRS. This project builds upon their findings by narrowing the focus specifically to the E1 Climate Change standard, aiming to provide a deeper, more specialized alignment tool that maps operational data to the precise Disclosure Requirements identified in their research.

3.2 Unstructured Narratives to Structured Insights

Bronzini et al. (2024) address the underlying challenge of ”greenwashing” by using LLMs to convert unstructured sustainability narratives into verifiable, structured data [3]. Their ”Glitter or Gold?” framework utilizes RAG to map qualitative corporate claims to specific regulatory categories, revealing that many sustainability efforts are obscured during

documentation. Bronzini’s work is foundational to this project’s PDF parser, or ”Document Chopper”; it proves that the extraction of information from text heavy PDFs must be grounded in a structured semantic framework to be actionable. This project adopts their approach to information extraction but extends it by incorporating coordinate-based meta-data (page numbers and headers) with the goal of eventual ”audit-grade” traceability that pure narrative extraction lacks.

3.3 Computational Efficiency and Sustainable AI

Literature has also begun to prioritize the environmental impact of the AI models themselves. Alevizos et al. (2025) introduced the concept of ”Sustainable Swarm Intelligence,” evaluating the carbon footprint of optimization algorithms and proposing an ”Emission Impact Metric” to link algorithmic choices to verified emission outcomes [1]. Their findings suggest a significant efficiency gap between different computational methods. This idea is echoed by Kozhipuram et al. (2025), who demonstrate that smaller, RAG-enabled models can achieve parity with massive 175B+ parameter models on knowledge intensive tasks while consuming a fraction of the energy [8]. By selecting a local, 7B-parameter Mistral model and an efficient 384-dimensional embedding space, this project aligns with the ”Green AI” frameworks proposed by Alevizos, ensuring that the tool used to monitor climate compliance is itself efficient and sustainably responsible.

4 Methodology

Dijkstra’s algorithm was chosen for single-source shortest path computation on graphs with non-negative edge weights, following the implementation presented in CLRS [4].

4.1 Data Ingestion and Pre-processing

I built a Data Ingestion Pipeline Using PyMuPDF and Regex to ingest and structure the 320 pages+ of legal documentation into a json file that can

4.2 Vector Database Implementation

A persistent ChromaDB instance serves as the system’s non-parametric memory, storing document chunks as high-dimensional vectors. Using the all-MiniLM-L6-v2 model, the database indexes 384-dimensional embeddings that allow for efficient similarity searches while maintaining a lightweight local storage footprint.

4.3 The Retrieval Pipeline

By implementing a k-nearest neighbors (k-NN) approach, the pipeline identifies and extracts the top- k most relevant context windows from the ESRS E1 standard to serve as factual grounding for the model.

4.4 The Generation Component

The generation layer utilizes a localized model to synthesize the retrieved legal segments into a coherent response. The model is constrained by a system prompt that enforces strict grounding, requiring the generator to cite specific Disclosure Requirements and preventing the hallucination of non-existent regulatory mandates.

4.5 User Interface Implementation

4.6 Accuracy Mapping of the Model

5 Results

An exemplary figure is shown in Figure 1. Any time a figure is used, it must have a caption and be directly referred to and discussed in the text.

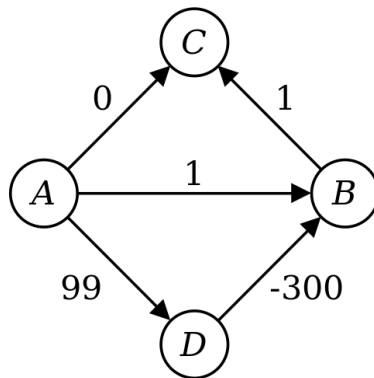


Figure 1: Dijkstra’s algorithm does not work on graphs with negative edge weights, as evidenced here.

5.1 Performance Metrics

5.2 Case Study: ESRS E1-1 Alignment

5.3 Sustainability Analysis: Local vs. Cloud Execution

6 Discussion and Future Work

6.1 Technical Limitations

6.2 Human Validation

6.3 Scalability

7 Conclusion

References

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